CAPSTONE PROJECT 2

Final Report

Consumer Adoption of Smart Home Security Devices in Urban Malaysia

by

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Abstract

Although the smart home security devices have experienced rapid growth globally, their adoption remains relatively low in Malaysia. This study examines the key barriers and motivations that shape consumers' intention to adopt smart home security devices in urban Malaysia. A research model was proposed by adding the constructs trust, privacy risk and security risk to the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) model. To validate the research model, Data collected from 244 respondents was evaluated using Partial Least Squares Structural Equation Modeling (PLS-SEM). These respondents reside in urban areas of Malaysia, specifically Kuala Lumpur and the Petaling district, and have at least a vague idea of smart home security devices. The results reveal that consumers' trust in adopting smart home security devices is positively impacted by privacy risk, behavioral intention is significantly and favorably influenced by trust, facilitating conditions, habit and social influence. The results also reveal that the actual use of smart home security devices is impacted by facilitating conditions, habit and behavioral intention. Overall, the findings contribute theoretical significance to the technology management literature and provide practical implications for companies as well as businesses to design solutions that improve the adoption of smart home security devices in urban Malaysia.

Keywords: Smart Home Security Devices, Adoption, Behavioral Intention, Trust, Security Risk, Privacy Risk.

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1.0 Introduction

According to Aldrich (2003), smart home technology is a house equipped with computing and information technologies that function to predict as well as respond to residents' needs. These smart home security technologies can provide residents with convenience and security through managing the home technologies through an application. Smart home technology is part of Internet of Things (IoT) that includes connected devices like sensors, monitors and so on. This project focuses on studying the consumer adoption of smart home security devices in Malaysia. Smart home security systems functions through utilizing a network of interconnected devices to protect user's home. The devices are connected to a central hub and it allows users to monitor and control the system through an application. The examples of smart home security devices include smart door lock, cameras, sensors, alarms and so on.

In recent years, smart home security devices are gaining global popularity due to the rapid development in IoT. SNS Insider (2025) reported that the global market size of smart home security has reached USD 74.42 billion in 2023, whereas it is forecasted to reach USD 169.63 billion by 2032. Statista Market Insights (2024) mentioned that the revenue of smart home security devices in 2024 Malaysia is USD 56.9 million. By 2029, it has forecasted to reach USD 105.9 million, which indicates that there is an increasing interest in the local market. This growing trend aligns with the Malaysia Smart City Framework (MSCF), which is published in 2018 by the Ministry of Housing and Local Government. This framework serves as a guide and reference to local authorities in planning and developing smart cities in Malaysia. The main pillars of competitive economy, enhanced quality of life as well as sustainable environment is achieved by this framework to address the urban issues and challenges in Malaysia. Government of Malaysia emphasizes that integrating smart solutions into urban planning, development and management can provide solutions to urban challenges and improve quality life of the community. Smart home security devices directly contribute to the vision of "Smart Living" outlined in the framework, where technology is leveraged to enhance safety and security of urban residents.

However, Malaysia's adoption rate of smart home technologies remains critically low as compared to other selected countries in Asia. According to Statista Market Insights (2024), Malaysia has a low generated revenue of smart home devices of 353.8 million USD against 32372 million USD, 556.5 million USD, 306.4 million USD for China, Indonesia and Philippines respectively. Based on Mokhtar and Ismail (2018), this might be due to the low

awareness of users towards the features and benefits provided by smart home security devices. He also mentioned that the smart home development in Malaysia is still new and most of the consumers have limited knowledge due to barriers like privacy and security data, cost, reliability and so on. Besides, academic studies related to adoption of smart home security devices are mostly carried out in rapidly developing countries. There are existing studies on smart home technologies in Malaysia, but few specifically focusing on smart home security devices.

1.1 Problem Statement

Recently, the demand for smart home technologies is continuously growing on the Internet of Things domain. Statista Market Insights (2024) shows that from 2019 to 2023, there has been an increase in the percentage of Malaysian adults which owns smart home devices. However, the slow adoption rate of smart home technologies in Malaysia compared to other Asian countries indicates that the adoption of smart home security devices is also relatively low. This is because the development of smart home security devices in Malaysia is still emerging, users may have limited awareness and knowledge about smart home security devices.

Moreover, the factors which influence consumer adoption of smart home security devices remains uncertain especially in urban Malaysia. Most of the other researchers focus on technical areas like security threats, development and usage of smart home technology. Besides, most of the studies are based in places like China, Jordan or Asian countries, with limited focus on urban Malaysian consumers. Few studies have examined the barriers and motivations which influence the consumer's intention to adopt smart home security devices. Pal et al. (2019) mentioned that consumers express their concerns about security and privacy, perceive usefulness, service cost, complexity of installation and use and so on. These concerns might be main factors that influence the consumer adoption of these technologies. By understanding the factors which influence consumers in adopting smart home security devices, including the identification of barriers and motivations help developers to design smart home security solutions which meet the needs of consumers, it can help to improve the user adoption rate of smart home security devices and safety of urban Malaysia.

1.2 Project Objectives

The project focuses on exploring the key factors that influence consumer's intention to adopt smart home security devices in urban Malaysia. It will outline both motivations like perceived usefulness and perceived ease of use. It will also outline barriers like privacy concerns and perceived cost. Through understanding the motivations and barriers of the consumer adoption of smart home devices, developers can carry out improvements and design features that meet consumer's needs on smart home security devices. This can increase the adoption rate and enhance residential safety in Malaysia's urban communities. The Unified Theory of Acceptance and Use of Technology 2 Model (UTAUT2) is tailored to better understand and explain the consumers' adoption of technology, specifically smart home security devices. The model will also be further extended by incorporating additional construct, like trust, security risk and privacy risk.

The objectives of this project are:

1. To identify the key barriers and motivations influencing the urban Malaysian consumers' intention to adopt smart home security devices by applying the extended Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) model.
2. To validate the proposed model using Partial Least Squares Structural Equation Modelling (PLS-SEM) through the SmartPLS 4 software.
3. To provide recommendations for businesses and companies on strategies to improve the adoption of smart home security devices among urban Malaysian communities.

1.3 Scope of Work

This study discovers the key barriers and motivations that influence consumer adoption in urban areas of Malaysia, particularly Kuala Lumpur and Petaling district in Selangor. These two regions are selected because they have a higher concentration of people, infrastructure and smart device availability as compared to rural areas. Besides, the target audience of this study is limited to Malaysians aged 17 years and above, they must have a basic knowledge or awareness on smart home security devices to answer the survey questions. Data will be collected through convenience sampling using an online Google Form survey and snowball sampling will be implemented to speed up the process of data collection. Moreover, this study has proposed the extended UTAUT2 model by including the additional constructs like trust,

privacy risk and security risk. The effect of the original UTAUT2 constructs on behavioral intention and actual use of smart home security devices will be evaluated. Partial Least Square Structural Equation Modeling (PLS-SEM) will be implemented to analyze the data collected using SmartPLS4. This includes measurement model assessment and structural model assessment. Furthermore, this study will implement a Likert-scale to develop a structured questionnaire that includes respondents' demographic questions and measurement items for relevant constructs. Pilot testing will be performed to check the validity and clarity of the questionnaire.

However, rural areas Malaysians are excluded in this study because these areas have limited accessibility to internet infrastructure and smart home security technologies. Smart home devices that are unrelated to home security such as smart lightning, fridges and home assistants will not be studied. Individuals under the age of 17 are not considered in this study because of ethical concerns and limitations in consent procedures. Moreover, the study is localized in urban Malaysia to understand specific cultural and behavioral factors, so it does not compare adoption rates across other countries although global studies are reviewed in this literature. This study will be based on data collected during a one time period, it does not track consumer's behavior in adopting smart home security devices over time as well as after actual outcome stage.

2.0 Literature Review

2.1 Smart Home Technology

Smart home technology is defined as a home setup where users can control appliances and devices anywhere using a mobile application connected to the internet. These technologies allow users to control the temperature, appliances, security devices and so on through mobile applications or voice-control assistants like Amazon's Alexa or Apple Siri. Since the consumer adoption of smart home technologies is rapidly increasing nowadays, a lot of the research explored the adoption of smart home technologies from various theoretical and practical perspectives to better understand the factors that influence user acceptance and behavior.

A study by Basarir-Ozel et. al (2022) has explored the factors that contribute to the adoption of smart home technologies. The authors mentioned that the existing literature still lacks studies that consider the future adoption of smart homes from a business perspective. Hence, three theoretical models, Diffusion of Innovation (DOI) theory, Unified Technology of Acceptance and Use of Technology 2 (UTAUT2) and Perceived Risk Theory (PRT) have been combined as an integrative model to determine the motivators and barriers of smart home technology adoption in business context. Relative advantages, image, enjoyment, design and technology innovativeness are the constructs of the adoption of smart home technology proven by the study. High cost, lack of trialability, lack of observability, lack of a trusted brand, lack of facilitating conditions and support service, complexity and technology anxiety are proven to be the barriers of smart home technology adoption.

The second study is by Shin et al. (2018), which focuses on addressing the problem of slow adoption of smart home technologies than expected. Besides, few previous studies have analyzed smart home technologies especially from a user perspective. Hence, technology acceptance model (TAM) has been extended through adding the two constructs, compatibility and privacy to determine the factors that influence the diffusion of smart homes. The study found that perceived ease of use, compatibility and perceived usefulness positively influences the purchase intention of users. Besides, the study found that older people with high incomes are the main target audience for purchasing smart home technologies, which suggests implementing products or services that are convenient as they are still unfamiliar in using these technologies.

Furthermore, a study by Liu and Chou (2020) also explores the traits and technology literacy level of consumers that adopt smart home appliances. The technology acceptance model (TAM) has been extended by including the constructs innovation adoption and computer

self-efficacy to evaluate the consumer's complex mind processing and behavioral intention to adopt smart home appliances. The original TAM focuses on explaining the outer factors of influencing users' behavioral intention, but not the inner factors like innovation adoption. The findings prove that innovation adoption positively affects perceived usefulness and perceived ease of use. This indicates that consumers will pay more attention to innovative products if they have more innovative adoption tendency. Besides, self-efficacy is proven to positively influence perceived ease of use and perceived usefulness, which means that people who have more confidence in their abilities are more likely to think the innovative products are easy to use.

2.1.1 Smart Home Security Technology

Smart home security systems are among the most widely adopted smart home technologies, driven by consumer demand for real-time alerts, greater control and remote access to home monitoring. These technologies include smart locks, video surveillance systems, alarm systems and so on, which enable users to monitor and secure their homes through a mobile application. There are some studies that have investigated the adoption of smart home security technologies using different theoretical models.

One of the studies is by Golda et al. (2022), which aims to present an exploratory study on how people perceive the usefulness and privacy risks of smart video surveillance systems that utilize digital skeletons instead of facial or biometric data. This study has implemented TAM to investigate the differences between gender and privacy level in perceived privacy risk, as well as how perceived usefulness of smart video surveillance system is influenced by gender, location type and time of visit. The study proves that perceived usefulness varies significantly depending on the public area and time of day. For example, surveillance is seen as more useful during the day in places like schools and parks. At night, it's perceived as more useful in low-traffic or potentially dangerous areas like parking lots and sidewalks. The study also found that participants perceived different privacy risks based on the type of surveillance system. Systems that analyze only digital skeletons or crime-related objects are perceived as least risky. For systems that record video footage, but censor biometric features are considered to carry a moderate level of privacy risk. The highest level of concern is associated with systems that analyze biometric features such as facial recognition, because these features can identify individuals and are seen as most intrusive.

Moreover, another study that is related to the adoption of smart home security technology is by Bhupathy and Krishnan (2019). The TAM model was adopted to discover customer buying behavior towards the security surveillance system, especially among institutional buyers in Coimbatore, South India. Safety features of CCTV cameras and the factors affecting consumers' purchase intentions are studied by the authors in this research. Many organizations have recognized the growing importance of security because of the rising crime and theft, yet they have not fully adopted modern security technologies. Hence, this study would like to understand the needs and expectations of consumers towards surveillance products. The findings of the study show that brand, price and camera resolution are the potential parameters that influence customers to purchase surveillance cameras. Besides, crime reduction factor and safety factors are positively correlated, which indicates that customers might think that crime can be reduced when they feel protected while using the technologies.

Furthermore, the third related study is by Liu et al. (2024), where an extended model is proposed through combining Expectation Confirmation Model (ECM) and Unified Theory of Acceptance and Use of Technology (UTAUT2) to discover the factors that influence users' intention to continue using smart locks. The authors would like to examine whether the common adoption factors like brand, cost and social influence still matter in influencing the consumer's intention to continue using these technologies. The findings show that satisfaction and habits directly predict users' continuance intention, which means that people who derive satisfaction from experience tend to repeat the behavior. Results show that trust significantly impacts satisfaction, which indicates that when users have trust in a product, there's a high chance that they will accept and use it. The results also show that trust mediates the relationship between factors like security risk, reliability and brand.

2.2 Overview of Existing Models Used in Literature

Due to the rapid advancement of smart home technologies, understanding the factors that drive or hinder consumer adoption has become an important area of research. Researchers like Davis, Fishbein, Venkatesh and so on have turned to established theoretical models from the fields of information systems, psychology and behavioral sciences to effectively predict and explain user behavior towards adopting new technologies. These models provide structure frameworks to discover how a person's intention to adopt new technologies is affected by different psychological, social and technological factors. Technology Acceptance Model

(TAM), Theory of Reasoned Action (TRA), Theory of Planned Behavior (TPB), and the Unified Theory of Acceptance and Use of Technology (UTAUT and UTAUT2) are mostly widely adopted by studies to discover the factors that influence the adoption of smart home technologies. This section provides a review of these theoretical models, highlighting their core constructs, focus and moderators, as well as how they have been applied in previous studies.

2.2.1 Technology Acceptance Model (TAM)

The demand for smart home security devices is increasing rapidly nowadays, hence exploring the adoption patterns of these technologies has become a crucial step in evaluating the success of smart home security devices in the market. TAM is one of the most widely used frameworks in previous studies in explaining the adoption of smart home systems.

Davis (1987) proposed the TAM which describes how the variables like perceived usefulness and perceived ease of use are influenced by system design features, which also influences users' attitude towards using the technology and their actual usage behavior. By integrating TAM, the adoption and usage of technologies are usually explained and predicted. This is because TAM is designed to explore why users choose to adopt or reject information technology, as well as how the system's design features influence their acceptance.

The TAM consists of three motivational variables, which are attitude towards using a system, perceived usefulness and perceived ease of use. In order to determine whether an individual actually uses a particular system, the construct attitude towards using a system, which is known as the level to which a person evaluates the use of the system in the context of their work is considered. The perceived usefulness is known as the level to which a person thinks that his performance in completing tasks will be improved while using a particular system. Moreover, the extent to which a person thinks that he would be free of physical and mental effort while using a particular system is known as perceived ease of use. The study proposed that perceived usefulness and perceived ease of use is directly influenced by the design features of a system.

There are several studies that have applied TAM to discover the adoption of different smart home technologies. Roloff and Lang (2023) adapted the TAM to discover how privacy and trust issues affect Germans' adoption of smart home devices. The authors mentioned that TRA and TPB only consider the attitudinal component of behavioral intention, whereas TAM is more suitable in this study as it focuses on predicting the acceptance of new technologies. They have applied TAM as the structural backbone of their extended framework and included

the two factors, trust and privacy to enhance the predictive power for smart home technology adoption. The trust variable is a user's belief that a technology can perform the required tasks (functionality), can effectively assist the user during usage (helpfulness) and function in a consistent and dependable manner (reliability), which is commonly modelled as system trust. This study extends the TAM framework because the original TAM does not consider factors such as privacy concerns that influence user's intention to reject using these devices.

Another study by Park et al. (2017) also applied the TAM, which focuses on determining the main factors that influence users to adopt smart home services. The original TAM is extended in this study to include various factors such as integrated enjoyment, perceived connectedness, system reliability, control, compatibility and so on. TAM is applied in this study as it is a widely recognized theoretical framework, which is used to understand how user adopt and use a new technology. Besides, the study also addresses the problem of how security, comfort, hedonic and economic values are structurally related to adoption behavior. This study applies structural equation modelling (SEM) to analyze the survey data from Korean respondents. The findings of the study show that the users' intention to adopt smart home services is strongly predicted by perceived usefulness. Attitude has a positive impact whereas perceived cost has negative impact towards the usage intention.

2.2.2 Theory of Reasoned Action (TRA)

Theory of Reasoned Action (TRA) is another model that is often applied by studies to explain the adoption of a particular technology. Fishbein and Ajzen (1975) proposed the TRA framework which aims to discover how does a person intends to perform a particular behavior. The three most important components of TRA are subjective norms, attitudes and behavioral intentions. A person's perception of social pressure to involve in a particular behavior is known as subjective norm. Moreover, a person's individual's positive or negative evaluation of a particular behavior is known as attitude. The way a person intends to behave in response to attitudes and beliefs is known as behavioral intention. The two constructs, subjective norms and attitude influence the behavioral intention, which is the main motivator of behavior. The authors mentioned that although TRA provides an understanding on how people form intentions to act, it doesn't discuss the beliefs that are important for a specific behavior or the types of people or social groups that influencing intention.

George et al. (2021) has implemented TRA in analyzing the user adoption of smart home technology. This study extends the TRA through adding specific brief constructs like fear and privacy concerns which influence attitude, to determine how these constructs influence the

customer's decision to purchase smart home security devices. This study extends the TRA through adding specific brief constructs like fear and privacy concerns which influence attitude. The study has also included a control variable known as experience with burglary. The original TRA is extended because it does not include specific beliefs or direct paths from beliefs to intention, belief affects intention indirectly. The study found that attitude towards purchasing is influenced by the belief about fear, while concerns about privacy also contributed. Both the factors, individual's attitude and the views of important others have similar impact on influencing their intention to purchase. Moreover, individuals that have personal experience of burglary were more likely to intend to purchase smart home security devices.

The second study that applies Theory of Reason Action (TRA) is by Mamonov and Benbunan-Fich (2021). The authors would like to assess the predictive value of selected constructs to forecast users' intention to adopt smart locks as well as examine the differences in gender in these predictive relationships. The research gap of this study is that the older theoretical models don't work well when they are used to understanding the factors of people choosing to use Smart Home Technologies. To address the research gap, a three-phased mixed-methods study is carried out through applying TRA as a foundation to extract context-specific beliefs regarding the adoption of smart locks. The study found that the adoption intention for both women and men is only significantly influenced by the construct perceived relative advantage. Moreover, perceived novel benefits influences women whereas novel benefits do not significantly influence men, unless the smart lock worked better than regular locks.

2.2.3 Unified theory of acceptance and use of technology (UTAUT)

Another model that is commonly adopted by existing studies to explore the adoption of smart home technology is the Unified Theory of Acceptance and Use of Technology (UTAUT). Venkatesh et al. (2003) has developed UTAUT by including the constructs, which are derived from existing eight theoretical models. They are performance expectancy, effort expectancy, social influence and facilitating conditions. The level to which an individual believes that using a particular system will help them to achieve gains in job performance is known as performance expectancy. The relationship between performance expectancy and intention will be moderated by gender and age. Effort expectancy is defined as the degree of ease associated while using a particular system. Gender, age and experience are the moderators that impact a person's intention to use a system. Besides, the level to which a person thinks that others believe he or she should use the new system is known as social influence. The study proposed that moderators like gender, age, voluntariness and experience will impact the social influence on

behavioral intentions. The level to which a person thinks that the organizational and technological resources required are available to support the use of a system is known as facilitating conditions.

Kang et al. (2022) has applied UTAUT to explore the consumer adoption of smart home devices. This study focuses on exploring the factors and analyzing the behavior of users in adopting Smart Home Health Care Services (SHHS) in South Korea, through developing a research model by adopting UTAUT and Task-technology fit (TTF) theoretical models. Besides, the study incorporated potential relationships identified from previous research, such as link between TTF to performance expectancy, as well as TTF to behavioral intention and examined their validity. The authors linked performance expectancy with TTF, it shows that when SHHS functions aligns well with healthcare-related tasks, there's a high possibility users will think that the technology is useful. However, the study found that TTF and behavioral intention are negatively correlated with each other. This indicates that users still hesitate to adopt SHHS due to complexity or unfamiliarity although they recognize a good task-technology fit. Moreover, the study found out that the technology characteristics of SHHS lead to lower effort expectancy, which indicates that the perceived effort needed to use the technology can be reduced by improving the system.

Furthermore, UTAUT2 is an extended model of UTAUT which also explains the user adoption of smart home technologies. Venkatesh et al. (2012) proposed the UTAUT2 model to assist organizations in the consumer technology industry design and market their products more effectively to various demographic groups. It extends the original UTAUT model by including three additional key constructs, habit, price value and hedonic motivation to focus on consumer context. Hedonic motivation is known as the perceived fun or pleasure derived from using technology. Price value refers to consumers' cognitive trade-off between the perceived benefit of the applications and the monetary cost for using them. Habit is defined as the degree to which individuals tend to perform behaviors automatically. The proposed model shows that the behavioral intention is influenced by the constructs, hedonic motivation and price value, whereas both behavioral intention and behavior is impacted by habit.

Plohl and Babič (2024) extended the UTAUT2 model by including an additional construct, trust to explore the factors that influences behavioral intention and technology use. This study focuses on addressing the issue of UTAUT2's unclear explanatory power in the existing research on smart home technology adoption. The study mentioned behavioral intention can be significantly predicted by hedonic motivation and price value, whereas behavioral intention and actual use can be significantly predicted by habit. On the other hand,

social influence, effort expectancy as well as facilitating conditions showed mixed significance across review studies. This means that demographic factors or the complexity of the product may influence these constructs. The predictive power of UTAUT2 can be improved by integrating the trust factor because it significantly improved the explanatory power in several models.

2.2.4 Theory of Planned Behavior (TPB)

TPB is the fourth theoretical model that is commonly used by studies to explain why users intend to use these smart home technologies. Ajzen (1991) has proposed TPB to forecast and explain human behavior under certain situations. Besides, TPB is a framework often used to deal with complex human social behavior as it has incorporated social and behavioral sciences concepts. TPB is extended from Theory of Reasoned Action (TRA) to overcome the limitations of the original framework in describing behaviors which people have incomplete volitional control. TPB also mentioned that the variables attitude toward behavior, subjective norm and perceived behavioral control influences the intention, which then influences user behavior. Perceived behavioral control is a person's belief on how easy or difficult it is to perform a particular behavior, based on perceived internal and external factors.

A research paper by Ji and Chan (2019) has applied TPB as a backbone theory to develop a research model, aiming to explore the factors influencing resident's intentions to adopt smart home energy technology. The norm activation model is also applied in this study to account for moral and personal norms, like environmental concern and social responsibility. In this study, the construct of attitude is split into two dimensions, which are attitude towards technical performance and attitude towards economic performance. Besides, the links between the constructs, attitude towards technical and economic performance, social norm, perceived behavioral control and the intention of residents in Guangdong to adopt Smart Home Energy Technology is explored in this study. The authors found that social norm, technical performance, perceived behavior control and personal norm is positively related to the adoption intention of Smart Home Energy Technology. Moreover, adoption intention is not positively influenced by the variable attitude towards economic performance.

Furthermore, a study by Yang et al. (2017) also applied TPB into their study to investigate the motivators and barriers of customer's intention in adopting and using smart home services by proposing an extended model on TPB. This proposed model not only added the constructs service usability, but also negative influences and level of trust in service providers like risk concern factors. The six exogenous variables that were proposed are trust in

service providers, security risk and physical risk, mobility, automation and interoperability. The authors demonstrated how TPB may be used to predict the behavioral intentions of smart home service users. The study proven that the constructs attitude, subjective norm and perceived behavioral control are positively correlated with intention of usage. Moreover, the findings of the study showed that the proposed framework is useful in understanding the smart home services adoption through combining the TBP and additional proposed constructs.

2.2.5 Comparison table

Theory/Model	**Constructs**	**Moderators**	**Focus**	**Application Context**
Technology of Acceptance Model (TAM)	1.Perceieved usefulness 2.Perceived ease of use	1.Experience 2.Voluntariness	Explains how consumers' behavioral intention to use technology is influenced by its perceived usefulness and ease of use	Smart Home Devices Roloff and Lang (2023), Smart Home Services Park et al. (2017).
Theory of Reasoned Action (TRA)	1.Attitude toward behavior 2.Subjective norm	1.Experience 2.Voluntariness	Explains behavioral intention based on attitude and subjective norms.	Smart Home Security Devices George et al. (2021), Smart Home Technologies Mamonov and Benbunan-Fich (2021).
Unified Theory of Acceptance and Use of	1.Performance expectancy	1.Gender 2.Age	A unified model to describe how	Smart Home Health Care

Technology (UTAUT)	2.Effort expectancy 3.Social Influence 4.Facilitating conditions	3.Experience 4.Voluntariness	individuals adopt and use technology, focuses on organizational context	Services Kang et al. (2022).
Unified Theory of Acceptance and Use of Technology 2 (UTAUT2)	1.Performance expectancy 2.Effort expectancy 3.Social Influence 4.Facilitating Conditions 5.Hedonic Motivation 6.Price Value 7.Habit	1.Gender 2.Age 3.Experience	Explains consumer technology adoption and focuses more on consumer context than organizational ones.	Smart Home Technology Plohl and Babič (2024).
Theory of Planned Behavior (TPB)	1.Attitude toward behavior 2.Subjective norm 3.Perceived behavioral control	1.Experience 2.Voluntariness 3.Gender 4.Age	Extends TRA through adding perceived behavioral control to better predict behavior under incomplete volitional control.	Smart Home Energy Technology Ji and Chan (2019), Smart Home Services Yang et al. (2017).

Table 2.2.5: Comparison of Theoretical Models Relevant to Smart Home Technologies

Table 2.2.5 presents a comparison of theoretical models in terms of their constructs, moderators, model focus and respective application contexts. The TAM model includes the constructs of perceived usefulness and perceived ease of use, with experience and voluntariness as moderators. This model explains how these two constructs influence consumers' behavioral intention to use a technology. The TRA model consists of the constructs attitude toward behavior and subjective norm, which together explain behavioral intention. It also includes experience and voluntariness as moderators. Moreover, the UTAUT model comprises the constructs performance expectancy, effort expectancy, social influence and facilitating conditions, along with the moderators gender, age, experience and voluntariness. This model aims to explain how individuals adopt and use technology within an organizational context. The UTAUT2 model is an extension of the UTAUT model, incorporating additional constructs such as hedonic motivation, price value and habit, while gender, age and experience as moderators. This model focuses on consumer technology adoption in a consumer context. Lastly, the TPB model includes the constructs attitude toward behavior, subjective norm and perceived behavioral control, with experience, voluntariness, gender and age as moderators. This model extends the TRA model to better predict individual behavior under conditions of incomplete volitional control.

2.3 Gaps in literature

Due to the increasing demand for smart home security technologies nowadays, understanding the factors, especially the drivers and barriers of consumer adoption is important. Although existing research has applied theoretical models such as TAM, TRA, TPB and UTAUT to explore the adoption of smart home technologies, we found a few gaps that remain in existing literature.

The first gap in existing research is the lack of studies that examine how barriers and risk perceptions affect the behavioral intention of users toward adopting smart security devices. Most of the studies have applied theoretical models like TAM and UTAUT, which focus heavily on positive adoption factors such as perceived usefulness, perceived ease of use, performance expectancy and so on. However, fewer studies have addressed barriers to adoption,

including concerns on cost issues or data breaches, which may influence user's rejection to adopt these technologies.

Next, the second gap in existing literature is that only a minority of the published studies focused specifically on the adoption of smart home security devices. Instead, most studies focused on the broader domain of smart home technologies without narrowing the scope to security-related applications. Many of these studies have explored areas like smart home healthcare services, energy management systems and so on. For example, a study by Ji and Chan (2019) have evaluated the factors that influenced residents' intention to adopt smart home energy technology in China.

Furthermore, the third gap in existing literature is most of the studies on smart home technology adoption have been conducted outside of Malaysia, such as Germany, South Korea and China. Few studies specifically explore how Malaysian consumers accept or reject smart home security devices. The adoption intentions may vary across countries due to differences in user behavior, privacy concerns and economic conditions. Thus, the findings from other countries cannot be directly applied to Malaysian urban consumers. Localized research should be conducted to investigate the motivations and barriers that influence Malaysian users in adopting smart home security devices.

2.4 Conclusion

In short, the literature review section has provided valuable insights on how different studies have applied different theoretical models to discover the factors that influence consumer adoption of smart home security technologies. The commonly used theoretical models like TAM, TRA, TPB, UTAUT and UTAUT2 help us to understand how consumers' intention to adopt new technology is impacted by social influence and perceived ease of use. However, I have identified several gaps in existing literature. The first is that existing literature lacks studies that examine how barriers and risk perceptions affect the behavioral intention of users toward adopting smart security devices. The second gap is that only a minority of the published studies focused specifically on the adoption of smart home security devices. The third gap is that few studies specifically explore how Malaysian consumers accept or reject smart home security devices. Thus, my study will address these gaps by integrating the extended UTAUT2 model to investigate the key motivations and barriers which influence consumer's intention to adopt smart home security devices in urban Malaysia. The findings of my study are expected to provide insights that helps to understand technology acceptance of

smart home security devices within the urban Malaysian context and offer practical insights for companies and businesses. They can better address customer needs and implement improvements to smart home security devices to foster greater adoption and enhance residential safety in Malaysian urban communities.

3.0 Research Methodology

3.1 Extended UTAUT2

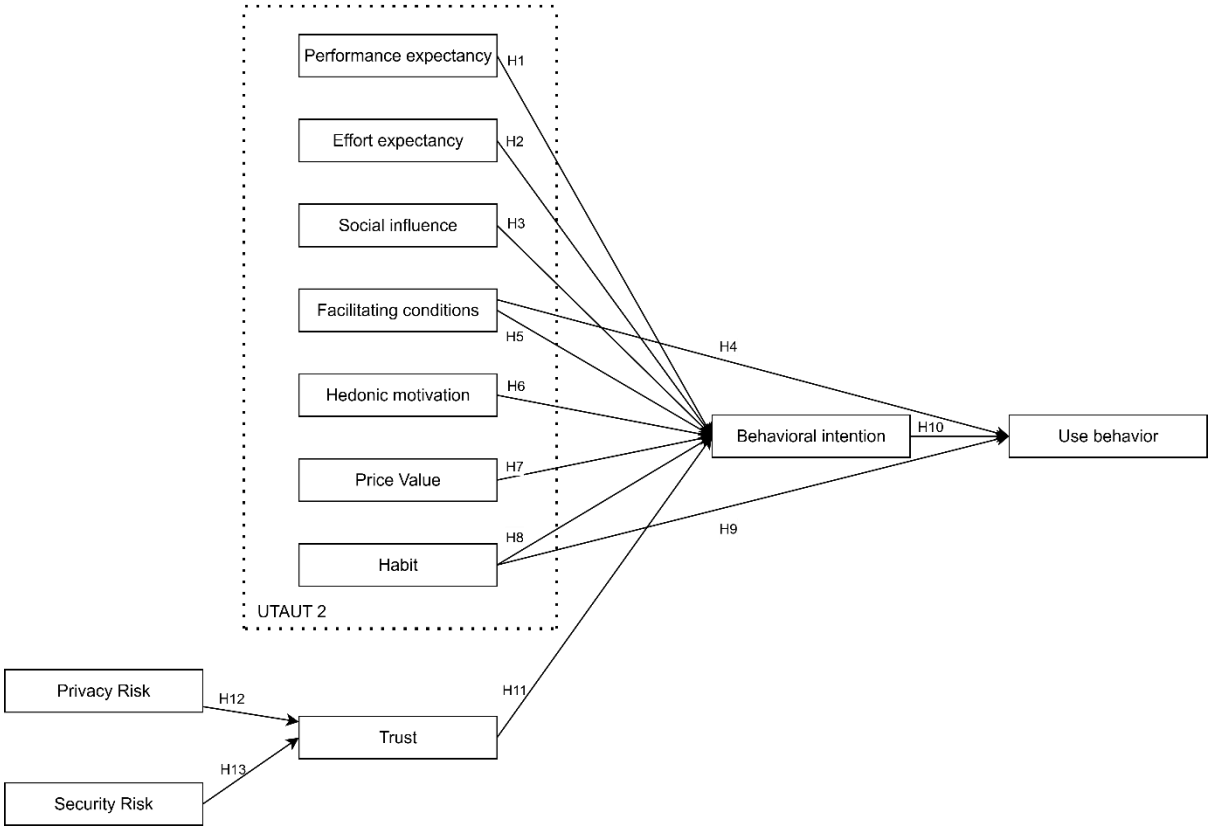

Figure 3.1: Extended UTAUT2

Figure 3.1 illustrates an extended UTAUT2 framework, incorporating privacy risk, security risk and trust as additional constructs. The framework analyzes the associations between the constructs and urban consumers' intention to use smart home security devices. The hypotheses, Performance expectancy (H1), effort expectancy (H2), social influence (H3), facilitating conditions (H5), hedonic motivation (H6), price value (H7), habit (H8) and trust (H11) are proposed to have direct effects on behavioral intention. It is hypothesized that behavioral intention has a direct effect on use behavior (H10). Direct relationships between facilitating conditions (H4) and habit (H9) and use behavior are proposed. Furthermore, privacy risk and security risk are incorporated as antecedents of trust. In short, this model aims to identify the barriers and motivational factors related to consumers' behavioral intention to adopt smart home security devices.

3.1.1 Performance Expectancy

Performance expectancy is the degree to which a user thinks that using a specific technology can help to meet his objective (Venkatesh et al., 2003). It is an important factor in determining whether specific technology will be adopted by users (Iqbal and Idrees, 2022). Based on previous studies, most findings show that users' behavioral intention to adopt smart home security technologies is significantly influenced by the variable, performance expectancy. In this study, performance expectancy is where an individual uses smart home security devices and believes that it will improve the security and protection of their homes, provide convenience through remote access and enhance their personal or family security needs. This study proposed that consumers' behavioral intention to adopt smart home security devices will be positively influenced by performance expectancy.

H1: The behavioral intention of consumers to adopt smart home security devices is positively influenced by performance expectancy.

3.1.2 Effort Expectancy

According to Venkatesh et al. (2003), effort expectancy is the level of comfort a user experiences when using a particular technology. Majority of the studies show that behavioral intention to adopt smart home technology is related to effort expectancy. In this study, it reflects how simple users can perceive the setup, operation and ongoing use of these systems. Users have high chance to adopt smart home security devices that are easy to learn and use as compared to complex ones. This study proposed that consumers' behavioral intention to adopt smart home security devices is positively influenced by effort expectancy.

H2: The behavioral intention of consumers to adopt smart home security devices is positively influenced by effort expectancy.

3.1.3 Social Influence

Social influence is the degree to which a user perceives that other people think it's important that they utilize a particular technology (Venkatesh et al., 2003). In this study, social influence refers to the extent where advice and opinions of people around an individual influences his decision to use smart home security devices. Since most people do not have in-

depth knowledge about smart home security devices, individuals are easily influenced by other users which then changes their behavioral intention (Liu et al., 2024). Social influence happens when an individual's behavior changes due to the pressure from others, often through physical, digital or social networks (Iqbal and Idrees, 2022). Majority of the studies proved that performance expectancy significantly influences the behavioral intention of users to use smart home devices. This study hypothesizes that social influence has a positive effect on consumers' behavioral intention to adopt smart home security devices.

H3: Consumers' behavioral intention to adopt smart home security devices is positively impacted by social influence.

3.1.4 Facilitating Conditions

Facilitating conditions is the degree to which users believe that they have the required technical resources and support to use the technology (Venkatesh et al, 2003). It also refers to an individual's belief that the necessary support and infrastructure are available to assist them in carrying out a specific behavior, especially when a problem arises (Iqbal and Idrees, 2022). In the smart home security devices context, this involves having necessary resources and knowledge to operate the devices, compatibility with other technologies in use and the availability of system support when issues occur. Kang et al. (2022) mentioned that an individual's intention to adopt smart home technology is directly influenced by facilitating conditions. The adoption of these technologies is also significantly influenced by behavioral intention as well as facilitating conditions. In this study, a hypothesis is proposed, where consumer's behavioral intention is positively influenced by facilitating conditions to adopt smart home security devices. Moreover, this study proposed that consumers' actual usage behavior of smart home security devices is positively influenced by facilitating conditions.

H4: Consumers' actual usage behavior of smart home security devices is positively influenced by facilitating conditions.

H5: Consumers' behavioral intention to adopt smart home security devices is positively influenced by facilitating conditions.

3.1.5 Hedonic Motivation

Hedonic motivation refers to the user's perceived enjoyment or pleasure when a specific technology is used. This variable was included to the extended UTAUT to cater the enjoyment dimension related to the usage of a system (Venkatesh et al., 2012). Perceived enjoyment can influence a user's decision to purchase a product. Users who enjoy and admire the usage of technology are more likely to adopt it as compared to others who do not (Iqbal and Idrees. 2022). In the smart home security devices context, users may enjoy using interactive mobile apps, voice-controlled features or smart automation routines of the devices, as well as willing to recommend them to others. Most of the studies found that hedonic motivation and behavioral intentions are significantly correlated (Plohl and Babič, 2024). Thus, this study proposes that hedonic motivation positively influences consumers' behavioral intention to adopt smart home security devices.

H6: Consumers' behavioral intention to adopt smart home security devices is positively influenced by hedonic motivation.

3.1.6 Price Value

Price value is a person's evaluation of whether the advantages of using a particular technology are worth the financial cost. Price value is considered positive when the perceived advantage of utilizing the technology is greater than its cost. (Venkatesh et al., 2012). A study found that the significant construct that influences behavioral intention is price value (Plohl and Babič, 2024). Besides, the user's decision to adopt new technologies may be influenced by the cost of maintenance as well as hardware and software costs (Iqbal and Idrees, 2022). In the smart home security devices context, price value reflects whether users think that the devices are reasonably priced, affordable and offering good value for money based on their perceived benefits. This study proposes that consumers' behavioral intention to adopt smart home security devices is positively influenced by price value.

H7: Consumers' behavioral intention to adopt smart home security devices is positively influenced by price value.

3.1.7 Habit

Habit is the degree to which a behavior is performed automatically by an individual because of experience and repetition (Venkatesh, 2012). Habit is usually developed through repeated actions over time based on past experiences. Once a habit is formed, individuals often repeat their behavior automatically in similar situations without conscious thought. As the behavior becomes a routine, individuals are driven by their subconscious mind to perform the behavior instead of making complex decisions based on satisfaction or the intention to keep using the technology (Liu et al., 2024). In this study, habit is reflected when individuals use the devices automatically and routinely in common situations, such as checking the smart doorbell when someone approaches the door or immediately opening the app to view camera footage when a motion alert is received. This study proposes the consumers' behavioral intention to adopt smart home security devices is positively influenced by habit. Besides, the study also proposes that consumers' actual usage behavior of smart home security devices is positively influenced by habit.

H8: Consumer's behavioral intention to adopt smart home security devices is positively impacted by habit.
H9: Consumers' actual usage behavior of smart home security devices is positively impacted by habit.

3.1.8 Behavioral Intention

Behavioral intention is an individual's consistent response toward using a particular technology, which helps to predict the future consumption patterns. Intention represents a person's readiness to take specific actions (Iqbal and Idrees, 2022). According to the existing studies, behavioral intention significantly influences users' intention to use and adopt these new technologies (Venkatesh et al., 2003). In the smart home security devices context, a strong behavioral intention indicates that users have a high possibility of utilizing these devices in their daily life. This study proposes that there should be a positive correlation between behavioral intention and actual use of smart home security devices

H10: Behavioral Intention has a positive effect on the actual use behavior of smart home security devices.

3.1.9 Trust

Trust is the belief or confidence developed through a set of experiences, observations, interrelationships and expected outcomes. In the smart home security devices context, user's confidence in the reliability, security and privacy protection of these devices is referred to as trust (Liu et al., 2024). When consumers trust a particular technology, there's a possibility they will adopt it (Mashal and Shuhaiber, 2019). According to the existing studies, the significant factor influencing consumer behavior in the information systems field is trust (Yang et al., 2017). Hence, in this study, the additional construct trust is added to the extended UTAUT2 model. The model posits a positive relationship between trust and consumers' behavioral intention to adopt smart home security systems.

H11: Consumers' behavioral intention to adopt smart home security devices is positively impacted by trust.

3.1.10 Privacy Risk

Privacy risk is the possible threats or vulnerabilities that may compromise an individual's personal privacy while using a certain technology. In the smart home security devices context, privacy risk refers to concerns about the collection and security of data. To provide better service to users, smart home security devices will collect data on users' activities, preferences and usage patterns during their operation, which involves users' privacy. Privacy breaches might happen due to potential unauthorized access, use or exposure of user's personal information collected through these devices (Liu et al., 2024). Thus, privacy risk can be included as one of the concerns for consumers in adopting smart home security devices in urban Malaysia. The existing studies show that consumers' attitudes towards adopting smart home security devices are negatively impacted by their perception of privacy risk. This research suggest an inverse relationship between privacy risk and consumer trust in smart home security devices.

H12: Consumers' trust in smart home security devices is negatively impacted by privacy risk.

3.1.11 Security Risk

Security risk is an important construct in understanding the security concerns of users to ensure development, deployment and use of smart home security technologies. It refers to the potential technical vulnerabilities, management negligence or external threats which may result in unauthorized access or cyber-attacks. Besides, it may also lead to information leakage, system failures or various types of damage or loss (Pal et al., 2019). In the smart home security context, users are more concerned about the risks that could compromise the privacy and safety of their homes. Smart home security devices often handle sensitive data and control access to physical spaces, hence if data breach happens it may lead to serious consequences. This study proposes that consumers' trust in smart home security devices is negatively influenced by security risk.

H13: Consumers' trust in smart home security devices is negatively influenced by security risk.

3.2 Sampling design

As sampling methods, this study will implement convenience sampling to conduct the survey online through Google Forms, whereas snowball technique will be used to increase the response rate. Convenience sampling, commonly used in prior research, is a non-probability approach that involves selecting respondents based on ease of access and availability. Researchers can reach out to their participants with less effort than compared to other non-random sampling techniques. Golzar et al. (2022) highlights that data collection requires less time when samples are drawn from readily available target populations. However, the convenience sampling has a limitation in producing a less representative and generalized sample. To address this problem, snowball sampling technique will be applied in this study. First, the survey is distributed to those respondents who meet the study criteria, who are then encouraged to refer to others who also meet the criteria and may be willing to join. This process continues as new participants refer to more individuals, which captures and increases the chain of respondents through social networks (Parker et al., 2019).

This study aims to gather data from a sample size of 250 to 350 respondents over a period of 1.5 months, approximately 6 weeks, which is enough to be used in analysis by applying Partial Least Squares Structural Equation Modeling (PLS-SEM). Based on Kock and Hadaya (2018), PLS-SEM is suitable for small sample sizes as compared to Covariance-based Structural Equation Modeling (CB-SEM) because PLS-SEM prioritizes maximizing the explained variance of endogenous constructs whereas CB-SEM focuses on reproducing the covariance matrix which requires larger samples. The target audience comprises Urban Malaysian residents currently living in Kuala Lumpur and Petaling district in Selangor. Participants should be aged 17 and above and possess awareness or experience with smart home security devices. Rural residents without internet access, individuals under 17 years of age and those with no interest in smart home security devices will be excluded from the study. The survey will be distributed to the respondents through e-mail, Microsoft Teams or WhatsApp by sharing the Google Form link. Besides, participants were approached in person and asked to scan a QR code to access and complete the survey. To increase sample representativeness in this study, pilot testing will be conducted to identify the underrepresented groups, then focus additional recruitment on those demographics. Furthermore, social media like Instagram and LinkedIn will also be used to distribute the survey to improve the sample's demographic representativeness and enable it to be more generalizable to Malaysian urban residents.

3.3 Questionnaire

The survey questionnaire of this study consists of two sections, where the first section aims to collect demographic data of the respondents like gender, age, education, current residential location and personal monthly income, followed by questions related to smart-home security devices awareness. These questions help to exclude respondents who have no prior knowledge about smart-home security devices. The question asked is "Do you know what smart home security technologies are?". Participants can choose response options like "no idea", "vague idea", "general idea", "good idea" and "already using smart home security devices". Those who choose "no idea" will be eliminated and sent to the end of the survey. Those who choose any other option will proceed to the second section (Mashal and Shuhaiber, 2019). Questions about the measurement indicators of the study will be included in section two. It contains four measuring items for each construct like performance expectancy, effort

expectancy, social influence, facilitating conditions, hedonic motivation, price value, habit, behavioral intention, trust, privacy risk and security risks. The model's constructs are measured in this section using Five-point Likert scale with 1 represents "strongly disagree", 5 represents "strongly agree", and 3 represents "neutral score".

3.3.1 Demographic Questions

Characteristics		**Reference**
Gender	Male Female	Liu et al. (2024), Kang et al. (2022).
Age	17-20 21-30 31-40 41-50 51 and above	Kang et al. (2022), Liu et al. (2024).
Education	Pre-U Diploma Advanced Diploma Bachelor Postgraduate	Liu et al. (2024).
Current residential location	Puchong Subang Petaling Jaya Damansara Sungai Buloh Shah Alam	Kang et al. (2022).

	Kapar Kota Raja Kuala Lumpur None of the above	
Do you know what smart home security (IOT) devices are?	No idea Vague idea General idea Good idea Already using smart home security devices	Mashal and Shuhaiber, (2018).

3.3.2 Measurement Items

Construct	**Item No.**	**Questions**	**References**
Performance Expectancy (PE)	PE1	Smart home security devices are useful in my daily life.	Kang et al. (2022), Iqbal and Idrees (2022).
	PE2	I am efficient when I use smart home security technologies.	
	PE3	Smart home security devices help in raising awareness of potential threats by providing early alerts.	
	PE4	The use of smart home security devices provides time savings.	
	EE1	It is easy to use smart home security devices.	Kang et al. (2022), Iqbal and Idrees (2022).

Effort Expectancy (EE)	EE2	It is easy for me to learn how to use smart home security devices.	
	EE3	When using smart home security devices, I require the few steps possible to achieve my goals.	
	EE4	My interactions with smart home security devices are clear and understandable.	
Social Influence (SI)	SI1	People who are important to me think that I should use smart home security devices.	Kang et al. (2022), Iqbal and Idrees (2022).
	SI2	I should use smart home security devices according to those who influence my behavior.	
	SI3	People whose opinions that I respect prefer that I use smart home security devices.	
	SI4	Due to the impact of mass media, I am more likely to use smart home security devices.	
Facilitating Conditions (FC)	FC1	I have the necessary resources to use smart home security devices.	Kang et al. (2022), Iqbal and Idrees (2022).
	FC2	I have sufficient knowledge to use smart home security devices	
	FC3	Smart home security devices are compatible with other technologies or systems I use.	
	FC4	If I encounter problems using smart home security devices,	

		system support will be available to assist me.	
Hedonic Motivation (HM)	HM1	I find using smart home security devices to be fun.	Iqbal and Idrees (2022)
	HM2	I enjoy using smart home security devices.	
	HM3	Using smart home security devices is entertaining for me.	
	HM4	I would suggest using smart home security devices to my friends.	
Price Value (PV)	PV1	I believe the pricing of smart home security devices is fair.	Iqbal and Idrees (2022)
	PV2	I think smart home security devices offer good value for cost.	
	PV3	At the current price, smart home security devices provide are worth the investment.	
	PV4	I can afford the prices of smart home security devices.	
Habit (HT)	HT1	Using smart home security devices has become automatic to me.	Gu et al. (2019).
	HT2	It feels natural to me to use smart home security devices.	
	HT3	When I need to perform a specific task, using the smart home service is an obvious choice for me.	

	HT4	Using smart home security devices has become a daily routine.	
Behavioral Intention (BI)	BI1	I will always try to use smart home security devices in my daily life.	Kang et al. (2022), Iqbal and Idrees (2022).
	BI2	I plan to recommend smart home security devices to my friends or family members in the future.	
	BI3	I would like to use smart home security devices whenever I get the chance.	
	BI4	I am open to learning how to use smart home security devices effectively.	
Trust (TR)	TR1	I trust the reliability of smart home security devices.	Liu et al. (2024)
	TR2	I believe smart home security devices can be trusted.	
	TR3	I find smart home security devices to be competent.	
	TR4	I feel that I have control over smart home security devices.	
Privacy Risk (PR)	PR1	I am worried that smart home security devices might store my personal data.	Liu et al. (2024)
	PR2	I am concerned that data collected by smart home security devices could be sold.	

	PR3	I am worried that smart home security devices might be able to predict when I'm home or away.	
	PR4	I fear that a smart home security device could be used to steal personal information.	
Security Risk (SR)	SR1	I am concerned that unauthorized users can gain access to my home through these devices.	Liu et al. (2024)
	SR2	I am concerned that smart home security devices might have security vulnerabilities.	
	SR3	I am afraid that smart home security devices could be hacked.	
	SR4	I am concerned that hackers might access my data through smart home security devices.	
Use Behavior (UB)	UB1	I plan to use smart home security devices in the future to monitor and manage my home security.	Nikolopoulou et al. (2020)
	UB2	I am willing to use smart home security devices as a supporting tool to enhance my home safety.	
	UB3	I believe using smart home security devices will be a pleasant experience.	

	UB4	I intend to use smart home security devices frequently once I adopt them.	

3.4 Descriptive Analysis

The descriptive analysis will be performed to present a summary of the characteristics of respondents regarding smart home security devices and provide an overview for the sample profile. This study targets urban Malaysian residents aged 17 and above, specifically residing in Kuala Lumpur and Petaling district Selangor. To make sure that the target audience is eligible to fill out the questionnaire, they should have basic awareness or experience related to smart home security devices. The demographic characteristics that will be analyzed are gender, age, education, monthly personal income and current residential location. Besides, this section will include a summary of respondents' awareness of smart home security technology. These data will be presented using descriptive statistics like frequency counts and percentages. The results will also be interpreted using visual diagrams like pie charts and bar charts.

3.5 PLS-SEM

This study will use the SmartPLS4 software to perform partial least squares modeling (PLS-SEM) and analyze the data collected. PLS-SEM instead of covariance-based SEM (CB-SEM) is adopted to evaluate the 13 hypotheses and examine the framework, because it is more flexible, it requires a smaller sample size and can handle different data formats as well as large number of variables (Ji and Chan, 2019). Besides, PLS-SEM is a suitable alternative, especially when extending an existing structural theory, UTAUT2. PLS-SEM is more effective than CB-SEM when dealing with complex research models, as it can test both formative and reflective constructs (Kang et al., 2022). Hence, PLS-SEM will be applied to analyze the extended theoretical model.

3.5.1 Pilot Testing

In this study, pilot testing is first conducted before a full-scale survey to test and improve the research instruments. Pilot testing is a feasibility study which is used to pre-test a particular

research instrument. Researchers can identify the potential points of failure in the research project and highlights where research procedures might be ignored. It also determines whether the proposed methods are too complicated or inappropriate (Teijlingen and Hundley, 2002). The pilot test of this study begins with small focus groups discussions, which is a qualitative research method to identify key issues to be addressed in full-scale questionnaire survey. The constructs used in this study, like performance expectancy, effort expectancy, social influence, facilitating conditions, hedonic motivation, price value, habit, behavioral intention, trust, privacy risk and security risks will be refined. Next, the survey questionnaire will be tested on wording, question order or range of multiple-choice answers to ensure the items are understandable and relevant to the target audience. A final pilot testing round will be performed to evaluate the research process, which are the ways of distributing and collecting questionnaires. The questionnaire will be distributed via email, WhatsApp, Microsoft Teams or in-person using codes, so the feedback collected from pilot testing will help to identify weaknesses or potential problems in the distribution process.

3.5.1.1 Pilot Testing Measurement model

Construct	**Item**	**Convergent Validity**		**Internal Consistency Reliability**		
		Outer loadings	**Average Variance Extracted (AVE)**	**Cronbach's Alpha**	**Rho_A**	**Composite Reliability**
Behavioral Intention	**BI1 <- BI**	0.929	0.852	0.942	0.942	0.958
	BI2 <- BI	0.891				
	BI3 <- BI	0.938				
	BI4 <- BI	0.933				
Effort Expectancy	**EE1 <- EE**	0.956	0.814	0.926	1.063	0.945

	EE2 <- EE	0.945				
	EE3 <- EE	0.740				
	EE4 <- EE	0.948				
Facilitating Conditions	**FC1 <- FC**	0.670	0.320	0.264	0.317	0.567
	FC2 <- FC	0.638				
	FC3 <- FC	0.648				
	FC4 <- FC	-0.068				
Hedonic Motivation	**HM1 <- HM**	0.926	0.754	0.885	0.962	0.922
	HM2 <- HM	0.964				
	HM3 <- HM	0.566				
	HM4 <- HM	0.952				
Habit	**HT1 <- HT**	0.952	0.822	0.932	0.997	0.948
	HT2 <- HT	0.969				
	HT3 <- HT	0.773				
	HT4 <- HT	0.919				
Performance Expectancy	**PE1 <- PE**	0.764	0.552	0.729	0.799	0.829

	PE2 <- PE	0.691				
	PE3 <- PE	0.603				
	PE4 <- PE	0.885				
Privacy Risk	**PR1 <- PR**	0.700	0.836	0.931	0.993	0.952
	PR2 <- PR	0.992				
	PR3 <- PR	0.970				
	PR4 <- PR	0.964				
Price Value	**PV1 <- PV**	0.926	0.715	0.858	0.880	0.908
	PV2 <- PV	0.778				
	PV3 <- PV	0.979				
	PV4 <- PV	0.663				
Social Influence	**SI1 <- SI**	0.736	0.550	0.706	0.804	0.824
	SI2 <- SI	0.877				
	SI3 <- SI	0.816				
	SI4 <- SI	0.473				
Security Risk	**SR1 <- SR**	0.968	0.910	0.968	1.011	0.976
	SR2 <- SR	0.967				
	SR3 <- SR	0.895				

	SR4 <- SR	0.983				
Trust	**TR1 <- TR**	0.456	0.532	0.763	0.626	0.813
	TR2 <- TR	0.811				
	TR3 <- TR	0.780				
	TR4 <- TR	0.808				
User Behavior	**UB1 <- UB**	0.959	0.875	0.953	0.959	0.966
	UB2 <- UB	0.926				
	UB3 <- UB	0.946				
	UB4 <- UB	0.911				

Figure 3.5.1.2: Measurement Model Results

According to the pilot testing results, the outer loading values for the items in the constructs Behavioral Intention, Effort Expectancy, Habit, Security Risk and User behavior meet the minimum threshold of 0.7. For the constructs Trust, Social Influence, Price Value, Privacy Risk and Hedonic Motivation, there is one item in each construct that does not meet the threshold. Performance Expectancy has two items below the minimum threshold, whereas Facilitating Conditions has all items below the threshold. In addition, most constructs have an Average Variance Extracted (AVE) value exceeding the minimum threshold of 0.5, except for Facilitating Conditions, which has a low AVE value of 0.320.

In terms of internal consistency reliability, the Cronbach's alpha values for most constructs exceed the 0.70 threshold. However, Facilitating Conditions has a Cronbach's alpha of 0.264, which is below the threshold. For the Rho_A values, most constructs meet the 0.7 threshold, except for Facilitating Conditions and Trust, with values of 0.317 and 0.626

respectively. Regarding composite reliability, most constructs meet the minimum threshold, except for Facilitating Conditions, which has a composite reliability of 0.567.

In short, the pilot testing results indicate that most constructs in the measurement model show acceptable validity and reliability because majority of their items meet the required threshold. However, several constructs which are Hedonic Motivation, Performance Expectancy, Privacy Risk, Price Value, Social Influence and Trust contain one or two weak items that require refinement. This indicates that some items may not be strongly correlated with their respective constructs and should be revised. Most importantly, Facilitating Conditions shows poor measurement quality across all validity and reliability indicators, indicating that the items in this construct should be revised or replaced.

As for other results such as HTMT values, p-values, path coefficients and so on, these cannot be evaluated at this stage because a larger sample size is required.

3.5.1.2 Updated Measurement Items

Construct	**Item No.**	**Questions**	**References**
Performance Expectancy (PE)	PE1	Smart home security devices are useful in my daily life.	Kang et al. (2022), Iqbal and Idrees (2022).
	PE2	Use of smart home security devices makes me efficient.	
	PE3	Smart home security devices help in raising awareness of potential threats by providing early alerts.	
	PE4	Using smart home security devices helps me save time.	
	EE1	It is easy to use smart home security devices.	Kang et al. (2022), Iqbal and Idrees (2022).

Effort Expectancy (EE)	EE2	It is easy for me to learn how to use smart home security devices.	
	EE3	It takes only a few simple steps to use smart home security devices effectively.	
	EE4	Using smart home security devices is straightforward and understandable.	
Social Influence (SI)	SI1	People who are important to me think that I should use smart home security devices.	Kang et al. (2022), Iqbal and Idrees (2022).
	SI2	I should use smart home security devices according to those who influence my behavior.	
	SI3	People whose opinions that I value prefer that I use smart home security devices.	
	SI4	Due to the influence of mass media, I am more likely to use smart home security devices.	
Facilitating Conditions (FC)	FC1	I have the necessary resources to use smart home security devices.	Kang et al. (2022), Iqbal and Idrees (2022).
	FC2	I understand how to operate smart home security devices.	
	FC3	Smart home security devices are compatible with other technologies or systems I use.	
	FC4	If I have difficulty using smart home security devices, there will	

		be support from the system to help me.	
Hedonic Motivation (HM)	HM1	I find using smart home security devices to be fun.	Iqbal and Idrees (2022)
	HM2	I enjoy using smart home security devices.	
	HM3	Using smart home security devices is entertaining for me.	
	HM4	I would suggest using smart home security devices to my friends.	
Price Value (PV)	PV1	I believe the pricing of smart home security devices is fair.	Iqbal and Idrees (2022)
	PV2	I think smart home security devices offer good value for cost.	
	PV3	At the current price, smart home security devices provide are worth the investment.	
	PV4	I can afford the prices of smart home security devices.	
Habit (HT)	HT1	I use smart home security devices without having to think about it.	Gu et al. (2019).
	HT2	Using smart home security devices is natural to me.	
	HT3	When I need to secure my home, I usually use smart home security devices.	

	HT4	Using smart home security devices has become a daily routine.	
Behavioral Intention (BI)	BI1	I intend to use smart home security devices in my daily life.	Kang et al. (2022), Iqbal and Idrees (2022).
	BI2	I intend to recommend smart home security devices to my friends or family members in the future.	
	BI3	I would like to use smart home security devices whenever I get the chance.	
	BI4	I am willing to learn how to use smart home security devices effectively.	
Trust (TR)	TR1	I believe that smart home security devices are reliable.	Liu et al. (2024)
	TR2	I believe smart home security devices can be trusted.	
	TR3	I find smart home security devices to be competent.	
	TR4	I feel that smart home security devices are controllable.	
Privacy Risk (PR)	PR1	I am worried that smart home security devices might store my personal data.	Liu et al. (2024)
	PR2	I am concerned that data collected by smart home security devices could be sold.	

	PR3	I am worried that smart home security devices might be able to predict when I'm home or away.	
	PR4	I fear that a smart home security device could be used to steal personal information.	
Security Risk (SR)	SR1	I am concerned that unauthorized users can gain access to my home through these devices.	Liu et al. (2024)
	SR2	I am concerned that smart home security devices might have security vulnerabilities.	
	SR3	I am afraid that smart home security devices could be hacked.	
	SR4	I am concerned that hackers might access my data through smart home security devices.	
Use Behavior (UB)	UB1	I plan to use smart home security devices in the future to monitor and manage my home security.	Nikolopoulou et al. (2020)
	UB2	I am willing to use smart home security devices as a supporting tool to enhance my home safety.	
	UB3	I believe using smart home security devices will be a pleasant experience.	

	UB4	I intend to use smart home security devices frequently once I adopt them.	

3.5.2 Assessment of Measurement Model

The outer model in SEM, commonly known as the measurement model, explains how each latent construct is measure by its associated observed indicators in the proposed extended UTAUT2 model (Ji and Chan, 2019). Convergent and discriminant validity are used as key criteria in evaluating the measurement model. The convergent validity assesses the internal coherence of a construct's indicators, whereas discriminant validity evaluates the empirical uniqueness of each construct relative to others in the model (Hair et al., 2019).

To establish convergent validity, indicator loadings are examined against the minimum criterion of 0.70. Meeting this requirement implies that the construct captures more than 50% of the variance in its indicators, thereby supporting satisfactory item reliability (Hair et al., 2019). Secondly, internal consistency reliability is subsequently assessed through composite reliability, with values exceeding 0.70 considered good reliability (Kang et al., 2022). Hair et al. (2019) suggests that reliability values from 0.6 to 0.7 are sufficient for exploratory research, values between 0.7 and 0.9 represent good reliability, and values above 0.95 may indicate overlapping or redundant indicators. Cronbach's alpha is also applied to evaluate internal consistency reliability. However, it often produces lower values than composite reliability due to the equal weighting of items, making it a less accurate reliability indicator (Hair et al., 2019). According to Mamonov and Benbunan-Fich (2021), a Cronbach's alpha value greater than 0.7 meets the criterion for acceptable internal consistency. Furthermore, convergent validity is further established through the assessment of average variance extracted (AVE) values across all constructs. The AVE value should be at least 0.5 and above, which indicates that the construct accounts for a minimum of 50% of the variance of its items. The AVE is calculated by squaring the loading of each indicator on a construct and computing the mean value (Hair et al., 2019).

The assessment then proceeds to discriminant validity, which is evaluated using the heterotrait-monotrait (HTMT) ratio. HTMT is derived by taking the ratio of the geometric mean of within-construct item correlations to the mean of correlations across different constructs. The threshold value for HTMT is 0.9, where value greater than 0.9 indicates the

absence of discriminant validity. The constructs are too similar or overlapping in what they measure if the HTMT value of the two constructs is above 0.9.

3.5.3 Assessment of Structural Model

To evaluate the structural model, metrics such as coefficient of determination (R^2), blindfolding-based (Q^2) values and the relevance and significance of path relationships are applied. Collinearity is first assessed to ensure the robustness of regression results, in line with the recommendation of Hair et al. (2019). Collinearity across constructs in the model is assessed by applying inner VIF diagnostics. It should be below 5 to make sure that the structural model is acceptable without multicollinearity (Kang et al., 2022). Subsequently, the R^2 values associated with the endogenous constructs is evaluated. This measure captures the degree to which the variance of each endogenous construct can be explained by the exogenous constructs. R^2 ranges from 0 to 1, with values closer to 1 indicating stronger explanatory capability. The f^2 effect size is validated to explain how each independent construct contributes to the variance explained in the dependent construct. Usually, f^2 value is reported when the rankings from the path coefficients and f^2 is different, to explain the true contribution of each factor. As outline by Hair et al. (2019) effect sizes measured using f^2 are classified as small when exceeding 0.02, medium when above 0.15 and large when greater than 0.35.

Besides, the evaluation also incorporates the Q^2 value to determine the model's predictive relevance. Kang et al. (2022) mentioned that predictive relevance is established when the Q^2 values exceed zero. The Q^2 values are obtained using a blindfolding procedure, where some of the data points are temporarily removed then predicted using this model. The predicted values are compared with the original ones. The Q^2 value will be higher if the predicted values are close to the real ones, which indicate the proposed model has better predictive accuracy. Finally, the relevance of path coefficients and statistical significance is assessed. The direction and strength of links between constructs are shown by path coefficient, which ranges from -1 to +1. To test path coefficients' significance, bootstrapping method is applied. It is a statistical technique that resamples the data many times to estimate the accuracy of path coefficients (Hair et al., 2019). Besides, the statistical significance of the hypothesized relationships between constructs is examined through hypothesis testing, which involves evaluating the t-values and p-values obtained through the bootstrapping procedure to confirm or reject each hypothesis.

4.0 Results

4.1 Demographic Results

Demographic Profile (n = 270)			
		Frequency	**Percentage (%)**
Gender	Male	59	21.9
	Female	221	78.1
Age	17-20	26	9.6
	21-30	221	81.9
	31-40	15	5.6
	41-50	1	0.4
	51 and above	7	2.6
Level of Education	Pre-U	5	1.9
	Foundation	6	2.2
	Diploma	16	5.9
	Advanced Diploma	6	2.2
	Bachelor	215	79.6
	Postgraduate	18	6.7
	None of the above	4	1.5
Current Residential Location	Puchong	27	10
	Subang	43	15.9
	Petaling Jaya	34	12.6
	Damansara	18	6.7
	Sungai Buloh	2	0.7

	Shah Alam	26	9.6
	Kapar	1	0.4
	Kota Raja	2	0.7
	Kuala Lumpur	104	38.5
	None of the above	13	4.8
Do you know what smart home security (IOT) devices are?	No idea	9	3.3
	Vague idea	28	10.4
	General idea	108	40
	Good idea	97	35.9
	Already using smart home security devices	28	10.4

Table 4.1: Demographic Results

The final sample consisted of 270 respondents. The distribution shown in Figure 4.1 reveals a higher proportion of female respondents (78.1%) compared to male respondents (21.9%). The sample consists of individuals aged 21-30 (81.9%), while the remaining respondents were distributed across the 17-20 (9.6%), 31-40 (5.6%), 41-50 (0.4%) and 51 and above (2.6%) age categories. Most respondents reported holding a Bachelor's degree, representing 79.6% of the total sample. The remaining respondents held a Pre-university qualification at 1.9%, Foundation at 2.2%, Diploma at 5.9%, Advanced Diploma at 2.2%, Postgraduate degree at 6.7% and selected "none of the above" at 1.5%. For current residential location, most respondents live in Kuala Lumpur at 38.5%, followed by Subang at 15.9%, Petaling Jaya at 12.6%, Puchong at 10%, Shah Alam at 9.6%, Damansara at 6.7%, Sungai Buloh at 0.7%, Kota Raja at 0.7%, Kapar at 0.4%, and those who selected "none of the above" at 4.8%. For the question "Do you know what smart home security (IoT) devices are?", 40% had a general idea, 35.9% had a good idea, 10.4% had a vague idea, 10.4% were already using smart home security devices and 3.3% had no idea.

4.2. Descriptive Statistics

Construct	Name	Mean	Standard deviation	Excess kurtosis	Skewness
Performance Expectancy	**PE1**	3.561	1.261	-0.573	-0.732
	PE2	3.631	1.272	-0.674	-0.687
	PE3	3.668	1.281	-0.603	-0.726
	PE4	3.615	1.290	-0.576	-0.729
Effort Expectancy	**EE1**	3.484	1.335	-0.899	-0.573
	EE2	3.504	1.320	-0.761	-0.644
	EE3	3.549	1.347	-0.896	-0.607
	EE4	3.602	1.347	-0.796	-0.686
Social Influence	**SI1**	3.193	1.400	-1.231	-0.266
	SI2	3.209	1.426	-1.279	-0.296
	SI3	3.254	1.423	-1.271	-0.336
	SI4	3.443	1.406	-1.110	-0.501
Facilitating Conditions	**FC1**	3.275	1.371	-1.224	-0.285
	FC2	3.475	1.338	-1.015	-0.494
	FC3	3.426	1.361	-0.940	-0.571
	FC4	3.533	1.319	-0.984	-0.502
Hedonic Motivation	**HM1**	3.496	1.363	-0.933	-0.549
	HM2	3.586	1.332	-0.848	-0.599
	HM3	3.520	1.298	-0.805	-0.560
	HM4	3.725	1.297	-0.619	-0.759
Price Value	**PV1**	3.148	1.395	-1.260	-0.284
	PV2	3.369	1.319	-1.026	-0.446
	PV3	3.381	1.339	-1.011	-0.414
	PV4	3.242	1.294	-1.080	-0.264
Habit	**HT1**	3.098	1.378	-1.292	-0.093
	HT2	3.242	1.335	-1.104	-0.357
	HT3	3.385	1.393	-1.081	-0.475

	HT4	3.303	1.459	-1.293	-0.347
Privacy Risk	**PR1**	3.262	1.354	-1.233	-0.276
	PR2	3.291	1.349	-1.157	-0.371
	PR3	3.156	1.334	-1.247	-0.184
	PR4	3.291	1.409	-1.307	-0.280
Security Risk	**SR1**	3.324	1.402	-1.287	-0.304
	SR2	3.316	1.353	-1.129	-0.389
	SR3	3.459	1.395	-1.141	-0.467
	SR4	3.381	1.414	-1.233	-0.365
Trust	**TR1**	3.340	1.341	-1.018	-0.435
	TR2	3.344	1.326	-1.057	-0.409
	TR3	3.443	1.319	-0.901	-0.544
	TR4	3.488	1.323	-0.838	-0.588
Behavioral Intention	**BI1**	3.398	1.389	-1.087	-0.461
	BI2	3.430	1.343	-1.013	-0.464
	BI3	3.643	1.284	-0.667	-0.690
	BI4	3.721	1.301	-0.486	-0.830
Use Behaviour	**UB1**	3.492	1.410	-0.998	-0.601
	UB2	3.520	1.371	-0.956	-0.595
	UB3	3.557	1.315	-0.811	-0.642
	UB4	3.631	1.374	-0.908	-0.648

Figure 4.2: Descriptive Statistics Table

This research adopted a five-point Likert scale to measure responses, with values ranging between 1 and 5. Figure 4.2 shows that all constructs have a moderate to high mean values, ranging from 3.098 and 3.725. The results indicate that HT1 has smallest mean value (3.098), in contrast to HM4, which exhibited the highest mean score (3.725). The standard deviation results indicate moderate variability in respondents' responses, ranging from 1.261 to 1.459, suggesting that respondents' opinions are not uniform. Social Influence shows the highest variability, whereas Performance Expectancy has the lowest variability in responses. Each construct exhibits a negative excess kurtosis value, implying that the data are more evenly distributed rather than highly concentrated. The constructs Habit, Security Risk, Privacy Risk and Social Influence show the flattest distributions. Moreover, the skewness values of all

constructs range between -1 and +1, indicate that there is no extreme skewness. This also indicates that the data distributions are negatively skewed and that most respondents reflect positive perceptions.

4.3 Measurement Model

Construct	Item	Convergent Validity		Internal Consistency Reliability		
		Outer loadings	Average Variance Extracted (AVE)	Cronbach's Alpha	Rho_A	Composite Reliability
Behavioral Intention	BI1 <- BI	0.902	0.816	0.925	0.925	0.947
	BI2 <- BI	0.904				
	BI3 <- BI	0.922				
	BI4 <- BI	0.885				
Effort Expectancy	EE1 <- EE	0.885	0.795	0.914	0.915	0.939
	EE2 <- EE	0.911				
	EE3 <- EE	0.888				
	EE4 <- EE	0.882				
Facilitating Conditions	FC1 <- FC	0.883	0.802	0.918	0.919	0.942
	FC2 <- FC	0.900				

	FC3 <- FC	0.915				
	FC4 <- FC	0.884				
Hedonic Motivation	**HM1 <- HM**	0.913	0.816	0.925	0.926	0.947
	HM2 <- HM	0.915				
	HM3 <- HM	0.908				
	HM4 <- HM	0.877				
Habit	**HT1 <- HT**	0.889	0.824	0.929	0.931	0.949
	HT2 <- HT	0.926				
	HT3 <- HT	0.897				
	HT4 <- HT	0.920				
Performance Expectancy	**PE1 <- PE**	0.899	0.804	0.919	0.919	0.942
	PE2 <- PE	0.909				
	PE3 <- PE	0.880				
	PE4 <- PE	0.898				
Privacy Risk	**PR1 <- PR**	0.923	0.821	0.927	0.933	0.948
	PR2 <- PR	0.916				

	PR3 <- PR	0.863				
	PR4 <- PR	0.921				
Price Value	**PV1 <- PV**	0.927	0.825	0.929	0.930	0.950
	PV2 <- PV	0.905				
	PV3 <- PV	0.910				
	PV4 <- PV	0.892				
Social Influence	**SI1 <- SI**	0.913	0.826	0.930	0.930	0.950
	SI2 <- SI	0.907				
	SI3 <- SI	0.930				
	SI4 <- SI	0.886				
Security Risk	**SR1 <- SR**	0.909	0.864	0.947	0.948	0.962
	SR2 <- SR	0.942				
	SR3 <- SR	0.928				
	SR4 <- SR	0.938				
Trust	**TR1 <- TR**	0.923	0.856	0.944	0.945	0.960
	TR2 <- TR	0.932				
	TR3 <- TR	0.934				
	TR4 <- TR	0.913				

User Behavior	**UB1 <- UB**	0.927	0.852	0.942	0.942	0.958
	UB2 <- UB	0.934				
	UB3 <- UB	0.932				
	UB4 <- UB	0.898				

Figure 4.3.1: Measurement Model Results

	BI	**EE**	**FC**	**HM**	**HT**	**PE**	**PR**	**PV**	**SI**	**SR**	**TR**	**UB**
BI												
EE	0.790											
FC	0.833	0.744										
HM	0.826	0.814	0.804									
HT	0.837	0.736	0.832	0.777								
PE	0.785	0.817	0.732	0.842	0.743							
PR	0.539	0.471	0.529	0.447	0.457	0.451						
PV	0.807	0.744	0.806	0.780	0.816	0.749	0.538					
SI	0.816	0.721	0.736	0.736	0.805	0.698	0.576	0.780				
SR	0.548	0.529	0.587	0.509	0.526	0.508	0.788	0.528	0.529			
TR	0.782	0.727	0.680	0.803	0.645	0.788	0.542	0.738	0.770	0.438		
UB	0.829	0.776	0.780	0.829	0.775	0.813	0.546	0.806	0.776	0.564	0.806	

Figure 4.3.2: HTMT Value Results

The measurement model evaluation involved examining internal consistency reliability and establishing convergent and discriminant validity. To establish convergent validity, outer loadings and average variance extracted (AVE) were assessed against threshold values of 0.70 and 0.50 respectively. The results illustrated in figure 4.3.1 indicate that all measurement items met the outer loading threshold and each construct satisfied the AVE criterion. Measurement indicators demonstrated strong loadings, ranging from 0.882 to 0.942, alongside AVE values that varied between 0.795 to 0.864. The findings demonstrate that the criteria for establishing convergent validity are successfully met.

Internal consistency reliability was further examined through Cronbach's alpha, rho_A and composite reliability measures, where their values should be larger than 0.7 for good

reliability. All measurement constructs exhibited acceptable internal consistency reliability, with Cronbach's alpha, rho_A and composite reliability values greater than 0.70.

Furthermore, discriminant validity was established by applying the heterotrait-monotrait (HTMT) approach, where the constructs should exceed the threshold of 0.9. Based on figure 4.3.2, the discriminant validity was determined satisfactory with all construct values surpassing the 0.90 threshold. In short, the measurement model can be considered reliable and valid.

4.4 Structural Model

Hypothesis Path	VIF (Inner Model)	f-square	T statistics	Path coefficients	P values	Support
BI -> UB (H10)	3.081	0.175	4.547	0.431	0.000	Yes
EE -> BI (H2)	2.985	0.014	1.338	0.100	0.181	No
FC -> BI (H5)	3.263	0.045	2.499	0.186	0.012	Yes
FC -> UB (H4)	2.980	0.052	2.322	0.232	0.020	Yes
HM -> BI (H6)	3.978	0.010	1.109	0.097	0.267	No
HT -> BI (H8)	3.747	0.048	2.593	0.206	0.010	Yes
HT -> UB (H9)	3.091	0.042	2.390	0.213	0.017	Yes
PE -> BI (H1)	3.428	0.003	0.480	0.050	0.631	No
PR -> TR (H12)	2.197	0.123	5.176	0.447	0.000	Yes
PV -> BI (H7)	3.377	0.004	0.580	0.057	0.562	No

SI -> BI (H3)	3.219	0.032	2.041	0.156	0.041	Yes
SR -> TR (H13)	2.197	0.005	1.057	0.086	0.291	No
TR -> BI (H11)	3.276	0.033	2.115	0.160	0.035	Yes

Figure 4.4.1: Structural Model Results

	R-square	**R-square adjusted**	**Q square**
BI	0.764	0.756	0.720
TR	0.264	0.258	0.249
UB	0.656	0.652	0.649

Figure 4.4.2: R-square and Q-square Results

Evaluation of structural model focused on multicollinearity, effect size, coefficient of determination, predictive relevance and path significance. Inner VIF values were used to assess multicollinearity with a recommended threshold of less than 5. The results presented in Figure 4.4.1 indicate that all VIF values met this criterion, suggesting no multicollinearity issues.

Besides, the effect size between constructs is assessed based on f-square values, where values around 0.02 indicates low-level effect, 0.15 indicates middle-level and 0.35 indicates high-level effect. Results show that the effects of effort expectancy, hedonic motivation, performance expectancy, price value and trust on behavioral intention were considered weak, as their corresponding values did not meet the minimum threshold of 0.02. The relationship between behavioral intention and use behavior demonstrates a moderate effect size, as the corresponding value exceeds the 0.15 threshold. The remaining values are considered satisfactory as the exceeded the low-level effect size.

The coefficient of determination (R-square) values of endogenous variables was also evaluated, where a value around 0.25 indicates low-level explanation, 0.5 indicates middle-level and values over 0.75 indicates high level explanation. According to figure 4.4.2, the results show that all R-square values of the endogenous variables were at an acceptable level. Behavioral Intention has a value of 0.756, which shows high-level explanation.

The predictive relevance of the PLS-SEM model was assessed based on the Q-square value, where the endogenous variables' values should be larger than 0. The results showed that the Q-square values of all endogenous variables were above 0, which is at an acceptable level. Behavioral intention has the highest Q-square values of 0.720, whereas Trust has the lowest Q-square value of 0.249. In short, the structural model results are valid to determine the path coefficients and to test hypotheses.

4.5 Hypothesis Testing

The t-statistics, path coefficients and p-values displayed in Figure 4.4.1 are used to assess hypothesis significance, based on a predefined significance level of α=0.05.

4.5.1 Performance Expectancy

Performance expectancy demonstrates a positive relationship (path coefficient = 0.05) with behavioral intention. H1 is not supported since the relationship is statistically insignificant (p – value =0.631). Prior researches suggest that performance expectancy is a key determinant of users' intention to adopt smart home security devices. Iqbal and Idrees (2022) mentioned that individuals in Pakistan believed that adopting these technologies could enhance home security and improve energy efficiency in their daily lives. However, in this urban Malaysian context, the relationship was not significant because most young adults aged 21-30 do not perceive smart home security as a personal necessity. Many of them live in apartments with existing security systems or in shared accommodations where security is managed by the landlords. Thus, the perceived benefits of these technologies are not strong enough to drive their adoption intention.

4.5.2 Effort Expectancy

Although effort expectancy has a positive effect on behavioral intention (path coefficient = 0.10), the relationship is not statistically significant, leading to the rejection of H2 (p -value = 0.181). Earlier studies generally mentioned that effort expectancy is an important predictor of consumers' intention to use smart home security devices. Maswadi et al. (2022) demonstrated that effort expectancy is important in shaping the intention of older adults in

Saudi Arabia, who have limited experience in using advanced smart devices, therefore face greater difficulty when learning new systems. However, the relationship was not significant in the urban Malaysian context because most young adults already familiar in using smart home security devices. They are exposed to various smart devices and find it is easy to learn and operate smart home security devices. Hence, effort expectancy does not play a decisive role in shaping consumers' adoption intentions.

4.5.3 Social Influence

Social influence positively influences behavioral intention (path coefficient = 0.156), with the effect reaching statistical significance, confirming H3. Research conducted by Maswadi et al. (2022) revealed that elderly individuals' intention to use smart home technologies in Saudi Arabia was largely independent of social influences, due to the limited diffusion of such technologies at the time. Unlike previous findings, the relationship holds significant explanatory power in the urban Malaysian context. Based on the demographic results, only 10.4% of the respondents already using smart home security devices, indicating that these devices are not yet mainstream. Individuals heavily rely on experiences and recommendations of peers, family members and social media to reduce uncertainty and consider to adopt smart home security devices. Therefore, social influence emerges as a key determinant shaping consumers' adoption intentions.

4.5.4 Facilitating Conditions

A significant positive relationship (path coefficient = 0.232) is observed between facilitating conditions and use behavior, thereby supporting H4. Moreover, facilitating conditions also show positive impact (path coefficient = 0.186) on behavioral intention. H5 is supported as the p-value of 0.012 is smaller than the significance threshold. Both hypotheses reported are consistent with the past studies. For example, Iqbal and Idrees (2022) mentioned about consumers' intention in using smart home technologies are impacted by the facilitating conditions. In urban Malaysian context, facilitating conditions is a determining factor influencing behavioral intention as the availability of supporting infrastructure such as high-speed internet, IT maintenance and device compatibility provide essential infrastructure needed for consumers to adopt smart home security devices. If the infrastructure is not complete,

consumers might refuse to use these devices. Facilitating conditions is also a determining factor influencing use behavior because convenient access to local customer support enables consumers to quickly seek assistance and resolve issues when technical issues arise with their smart home security devices. If local customer support is not easily reachable, consumers may be discouraged from adopting smart home security devices.

4.5.5 Hedonic Motivation

Hedonic motivation shows a positive impact (path coefficient = 0.097) on behavioral intention. However, H6 is not supported as the p-value is 0.267, which does not meet the significance threshold, which is consistent with the past studies. For example, Iqbal and Idrees (2022) found that consumers intention in using smart home technology is not affected by hedonic motivation. This is because consumers in Pakistan prioritize privacy risks, costs and data security over enjoyment, where hedonic motivation is not the driving factor when consumers adopt smart home devices. The relationship is also not significant in urban Malaysian context as consumers do not associate smart home security devices with fun. These devices are mainly adopted for privacy and safety protection, unlike technologies such as VR which provide enjoyment.

4.5.6 Price Value

Price Value shows a positive impact (path coefficient = 0.057) on behavioral intention. H7 is not supported because the p-value is 0.562 and does not meet the significance threshold, which is consistent with the past studies. For example, Iqbal and Idrees (2022) reported that price value does not significantly influence behavioral intention as consumers in Pakistan perceived that smart home systems as high-priced technologies. Although smart home technologies may provide significant value, the cost still hinder individuals' intention to adopt them. In the urban Malaysian context, most respondents in this study have never adopted smart home security devices because only 10.4% are currently using them. These respondents are unable to evaluate whether the devices are worth the investment or offer good value. Besides, most respondents aged 21-30 have diverse income levels, such as students, fresh graduates or early-career professionals. This results in inconsistent views in terms of affordability, as some individuals can afford to adopt smart home security devices while others may not.

4.5.7 Habit

Habit shows a positive impact (path coefficient = 0.206) on behavioral intention. H8 is supported as the p-value is 0.01, which has met the significance threshold. Habit also shows a positive impact (path coefficient = 0.213) on use behavior. H9 is supported as the p-value is 0.017 and has met the significance threshold. Both hypotheses are found consistent, Tak et al. (2023) mentioned habit is the driving factor of smart home technology adoption as most California consumers are already familiar with using smart home HVAC systems as part of their daily routine. Consumers' previous experiences with these technologies create familiarity and automatic usage patterns, making it easier for them to adopt smart home HVAC systems. In the urban Malaysian context, consumer's habit significantly shapes their intention to use smart home security devices as residents are already familiar in using app-based interactions in their daily lives such as online shopping, parcel tracking and e-wallet transactions. Since smart home security devices can also be controlled through mobile applications, this existing familiarity with app usage strengthens their intention to use these devices.

4.5.8 Behavioral Intention

Behavioral intention shows a positive impact (path coefficient = 0.431) on use behavior. H10 is supported because the p-value has met the significance threshold, which is consistent as compared to the previous studies. Venkatesh (2003) stated consumer's intention strongly predicts use behavior and acts as the core determinant across all frameworks, which is validated in numerous studies. The study found that the individuals are unlikely to adopt the technologies if they do not intend to use it regardless of system qualities. In this study, behavioral intention significantly influences the use behavior of smart home security devices as most urban areas of Malaysia have sufficient infrastructure, such as access to installation services and high-speed internet, which supports consumers in turning their intentions into actual usage.

4.5.9 Trust

Trust shows a positive impact (path coefficient = 0.160) on behavioral intention. H11 is supported as the p-value of 0.035 is smaller than the significance threshold, which is consistent

to the previous studies. Liu et al. (2024) stated that trust significantly influences consumer's behavioral to adopt smart locks as these devices directly control access to their homes and ensure safety. Consumer's confidence in using smart home security devices depends on factors such as privacy protection, reliability and safety, which strengthens their trust in using these devices. This study shows that trust significantly influences the behavioral intention of consumers in urban Malaysia because smart home security devices collect sensitive personal and home data, and consumers are concerned about issues such as data breaches, hacking and privacy violations. If consumers do not trust the device, they will not intend to adopt it.

4.5.10 Privacy Risk

Privacy risk shows a positive impact (path coefficient = 0.447) on trust. H12 is supported as the p-value has met the significance threshold. It is not consistent to Liu et al. (2024) study, where the results indicate privacy risk is not the significant factor that impact trust. This is because consumers are not overly concerned about privacy risks related to smart locks and most of smart home manufacturers have implemented strong privacy protection measures such as advanced encryption, selective data sharing and options to turn off data collection. However, privacy risk shows a significant impact on trust in this study as urban Malaysians are highly concerned about unauthorized surveillance, misuse of personal information and hacking. Malaysia has experienced several major data leak incidents in the recent years, which increases the public awareness about data security. Thus, when consumers perceived a risk of data exposure, their trust in smart home security devices will be affected.

4.5.11 Security Risk

Security risk shows a positive impact (path coefficient = 0.086) on trust. H13 is not supported as the p-value is 0.291, which does not meet the significance threshold. It is not consistent with Liu et al. (2024), which reported that security risk significantly influences trust of consumers in adopting smart locks. The study indicated that security breaches are more serious and urgent than privacy risks and most respondents were concerned that smart locks may lead to unauthorized access that could result in actual break-ins. However, security risk does not significantly influence trust in the urban Malaysian context because most consumers believe that adopting smart home security devices from reputable brands, like Xiaomi or

Samsung ensures their security, as these brands are perceived to have strong cybersecurity measures and follow the protection standards. This may be also due to limited technical knowledge urban Malaysians have regarding security vulnerabilities.

5.0 Discussion

5.1 Theoretical Significance

Nowadays, the trend of users using smart home security devices are continuously emerging in Malaysia. Hence, more in depth study is needed in order to determine the main drivers or barriers that influence the intentions of urban Malaysian consumers to use these devices.

First, this study makes an initial contribution to technology management research by addressing smart home security device adoption within the under-researched context of Malaysia. It was found that existing studies focused on Internet of Things or home automation in countries such as China and Saudi Arabia, only a limited number have explored the behavioral patterns of Malaysian consumers. Moreover, previous studies often overlooked important factors such as privacy risk, security risk and trust, which are the potential determinants of successful adoption of smart security devices. By incorporating the previously overlooked constructs, the theoretical understanding of local adoption patterns will be improved, as well as the potential factors which contribute to the low adoption rate of these devices in urban Malaysia will be identified.

Secondly, a new theoretical framework has been proposed in this study through extending UTAUT2 model by incorporating the constructs privacy risk, security risk and trust. These extensions provide deeper explanatory power for understanding consumers' behavioral intention in using smart security devices. Questionnaires were distributed to 270 respondents living in urban Malaysia, specifically within the Petaling district and Kuala Lumpur, and the theoretical model was validated using PLS-SEM. The study results indicate the importance of privacy risk in influencing consumer trust and demonstrate that trust is important in influencing the consumers' intention to use these devices.

Furthermore, the security risk and privacy risk's influence on trust, and the trust's influence on consumers intention to use smart security devices is assessed in this study. The findings indicate that consumers' intention in using these devices is significantly determined by their trust. It is also found that privacy risk significantly influences consumer trust, whereas security risk does not have significant effect. In order to promote greater adoption, businesses should focus strengthening consumer trust by ensuring the protection of personal information and provide user-controlled privacy settings, which can enhance users' behavioral intention to adopt these devices.

5.2 Practical Implications

Based on the study's findings, a number of practical implications can be derived for companies, businesses, policymakers and developers within the smart home security device sector.

Results from the study highlight the significance of privacy risk in influencing consumers' trust in smart home security technologies. Companies can enhance consumer privacy through implementing secure authentication mechanisms that prevent illegal access to smart home security devices. This may involve the adoption of multi-factor authentication mechanisms and the enforcement of robust password policies. Companies can also adopt encryption technologies to secure data transmission and storage of user's information. Besides, companies can show their data management practices by stating the purpose of data collection and identifying parties who have access the data, so that consumers can better understand how their personal information is handled (Liu et al., 2024).

Lower perceived privacy risk is associated with increased consumer trust, which subsequently enhances adoption intention. The study further demonstrates that trust has a significant influence on behavioral intention toward smart home security devices. Gultom and Asvial (2020) mentioned that consumers will assess the security level of the smart home security devices when consumers are concerned about the risks associated. Therefore, companies should ensure strong security measures of smart home security devices to increase users' trust.

Furthermore, the findings reveal that facilitating conditions play an important role in shaping adoption decisions. Companies and businesses are recommended to ensure that customer service hotlines and technical support are easily reachable, providing troubleshooting assistance, installation services and maintenance for smart home security devices. To address concerns related to device complexity and resource demands, developers should prioritize the development of fully integrated smart home security systems, which are easy to deploy and use (Singh, 2023).

Moreover, habit is one of the driving factors that significantly influences consumers' behavioral intention to adopt smart home security devices. Developers should understand consumers' needs and design smart home security technologies with functions that can be easily integrated into daily life, while avoiding overly complex or redundant features. They should also ensure the responsiveness and stability of the smart home security devices, as these

devices must operate reliably in real time to protect users' homes (Liu et al., 2024). Companies are also suggested to include automated features in smart home security devices that operate in the background, which increases convenience and allows users to operate the technology effortlessly (Tak et al., 2023). This seamless interaction strengthens consumers' habitual use of the technology.

Lastly, the results further demonstrate the significant role of social influence in determining adoption intentions among urban Malaysian consumers. Companies should focus on including reference groups in their product advertisements to show how people similar to the target users adopt and benefit from smart home security devices (Palas et al., 2022). They should also utilize the electronic word-of-mouth by encouraging users to share their opinions, recommendations and experiences on purchasing website or social media (Iqbal and Idrees, 2022). The promotion of user-generated content and online feedback plays an important role in influencing prospective consumers' adoption patterns.

6.0 Conclusion

This study presented an extended research model grounded in the UTAUT2 framework by incorporating the constructs trust, privacy risk and security risk. The proposed model can be used to determine the key motivations and barriers that influence urban Malaysian consumers' intention to install smart home security devices. The proposed model was validated using PLS-SEM, revealing that trust, facilitating conditions, habit and social influence are key positive determinants of consumers' intention. Results reveal that facilitating conditions, habit and behavioral intention are significant predictors of actual usage of smart home security devices, while privacy risk positively influence consumers' trust towards these devices. The study's theoretical and practical contributions underscore the importance of improving data transparency and privacy protection to build consumer trust. Moreover, the habitual use can be reinforced by prioritizing seamless automation features when designing smart home security devices. Companies and businesses can improve facilitating conditions by providing technical support and designing fully integrated devices, while strengthening social influence by leveraging electronic word-of-mouth in marketing. This research advances the understanding of smart home security device adoption in Malaysia by providing empirical evidence from urban areas. Finally, future research can focus on exploring rural populations in Malaysia, adopting stratified sampling to obtain a more balanced sample and conducting comparative analyses to explore how adoption behaviors vary between different smart home security device types.

6.1 Limitations and Future Work

This study offers useful insights into consumers' adoption of smart home security devices in urban Malaysia. However, there are some issues that exist in this study needs to be explored and addressed.

Firstly, this study focuses on discovering the consumer adoption patterns of smart home security devices sector. However, the findings are limited to smart home security devices and do not explain consumers' behavioral intentions toward other smart home technologies. Hence, it is recommended that future studies can conduct a comparative analysis as different smart home devices serve different purposes which vary in terms of usefulness and functionality (Liu et al., 2024). This could also understand the consumers' perceptions and attitudes towards using different types of smart home devices.

Next, this study collected responses from a wide range of age groups. However, the number of respondents in each gender and age category is not consistent. 58.83% of the respondents are aged 21-30, which means the study cannot fully represent all age groups. Future studies are suggested to adopt stratified sampling to achieve a more balanced or proportionate representation across different respondent categories, reduce the risk of over-representation from a single age group and enhance the generalizability of the findings to the broader population.

Furthermore, the study focuses solely on urban areas in Malaysia, specifically the Petaling District and Kuala Lumpur, which have highest population density. The sample size is relatively limited, with 270 respondents, which may further restrict the applicability of the results. Consequently, the findings may not be representative of consumers in rural areas or regions outside the Petaling District and Kuala Lumpur. Future studies are encouraged to compare adoption patterns between urban and rural communities to discover how adoption barriers and influencing factors differ across different geographical contexts.

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